What is AI?
A Short Introduction for the LangChain Document Loader Exercises

1. THE CORE IDEA

Artificial Intelligence is the field of building systems that perform tasks which normally require human intelligence - recognising speech, understanding language, spotting patterns, making decisions, generating images or code.

The useful mental model: traditional software is rules written by a person. You tell the computer exactly what to do, step by step. AI flips this. You show the system many examples, and it derives the rules itself. Nobody writes "a cat has pointed ears and whiskers" - you show it a million labelled photos and it learns what separates cats from everything else.

2. THE LAYERS

AI is an umbrella term, and the words nest inside each other:

AI	The broad goal: machines doing intelligent-seeming things
Machine Learning	Systems that improve from data, not hand-written rules
Deep Learning	Machine learning using many-layered neural networks
Generative AI	Deep learning that produces new text, images, audio, code

Each is a subset of the one above it. Every large language model is generative AI, which is deep learning, which is machine learning, which is AI.

3. HOW LEARNING ACTUALLY HAPPENS

A model holds millions or billions of internal numbers called parameters, or weights. Training works like a feedback loop:

1. The model makes a prediction.
2. Its answer is compared against the correct one.
3. The gap is measured as an error.
4. The parameters nudge slightly to shrink that error.

Repeat this billions of times and the weights settle into something that generalises to examples the model has never seen. Nothing in the process resembles understanding in the human sense; it is optimisation against a scoring function, carried out at enormous scale.

4. WHY LANGUAGE MODELS WORK AT ALL

A large language model is trained on one deceptively simple task: given some text, predict what comes next. Do that well enough, across a large enough corpus, and a surprising amount of capability appears as a side effect - grammar, factual recall, translation, arithmetic, the ability to follow an instruction.

This is why prompt wording matters so much. You are not issuing a command to a program with fixed behaviour. You are steering a probability distribution. Small changes in phrasing move that distribution, sometimes a lot.

5. TWO THINGS WORTH KNOWING

AI is statistical, not certain. It predicts what is likely, not what is guaranteed true. This is exactly why a language model can produce a confident, fluent, completely wrong answer - a failure mode usually called hallucination. Fluency is not evidence of correctness.

Data quality decides everything. A model trained on biased or thin data learns those flaws faithfully, and reproduces them at scale. Garbage in, garbage out, with a very large multiplier.

6. WHERE RETRIEVAL COMES IN

Because a model only knows what was in its training data, it cannot answer questions about your private documents, or about anything that happened after training finished. Retrieval-Augmented Generation solves this by fetching relevant text first and passing it to the model as context.

That pipeline is exactly what the rest of these exercises build:

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  Documents -> Document Loader -> Text Splitter -> Embeddings
            -> Vector Store -> Retriever -> LLM -> Answer
```

This PDF exists as sample input for the document loader stage.